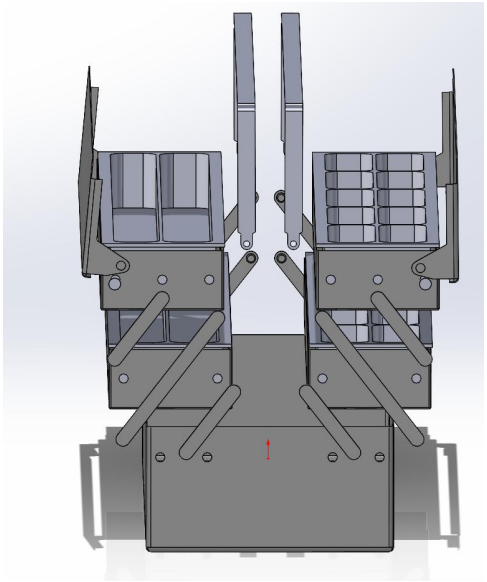
ISD 528 Team 17
Final Presentation

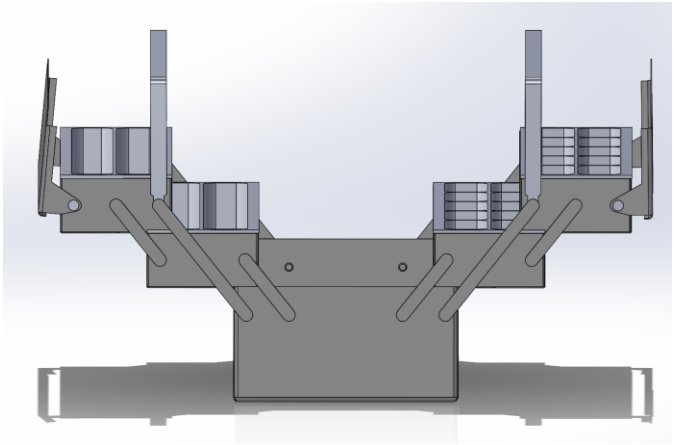
Skyler Do, Rachel Schillmoeller, Izzral Darwish Zulkifli, Ross Maynard, Walter LaMar, Jacob Rainey

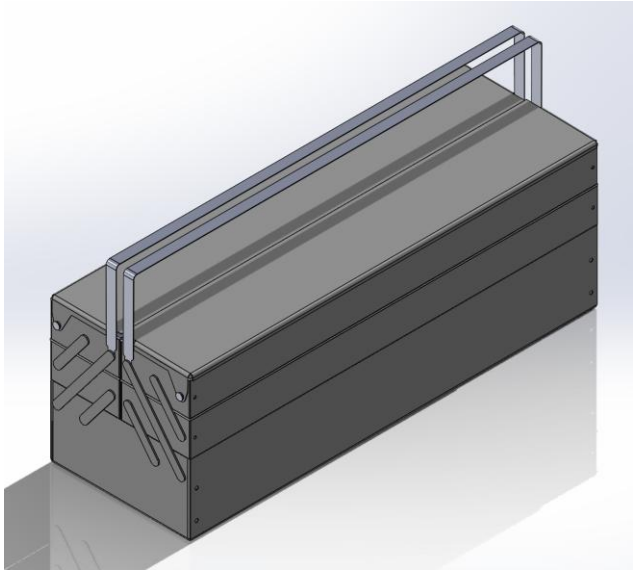
Product Description

Multi-tiered, collapsible toolbox with compartments



Exploded View





Production Volume: ~14,000 (per month) x 12 (months per year) = 168,000 per year

Source: *Amazon* sales numbers

Primary/Secondary Customers

Primary: “Blue collar” line of work (ex. mechanic, electrician, construction worker, plumber)

- Traditional toolbox use

Secondary: People who enjoy crafting,

- Store craft supplies such as beads, yarn, knitting needles, etc.







Product Analysis: FAST Diagram



Product Analysis: FCMM

	Functions								
	Extend Handle	Open Container	Organize Tools	Close Container	Resist Forces	Stabilize Containers	Assure Convenience	Adjust Configurations	Please Senses
Base					x	x			x
Small Link		x			x	x			x
Big Link		x			x	x			x
Supporting Tray			x					x	x
Divider A			x					x	x
Divider B			x					x	x
Flaps				x	x	x			x
Handles	x				x		x		x

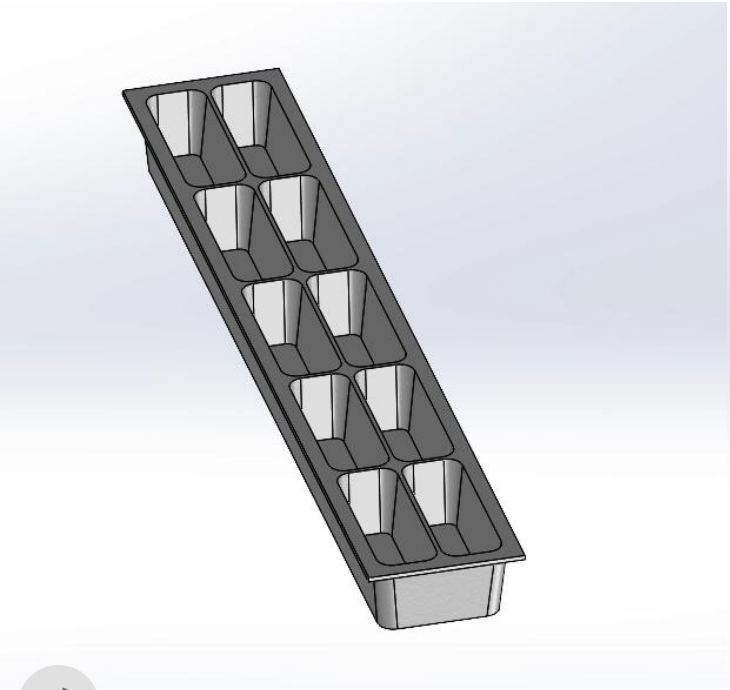
Product Family Design: CCT

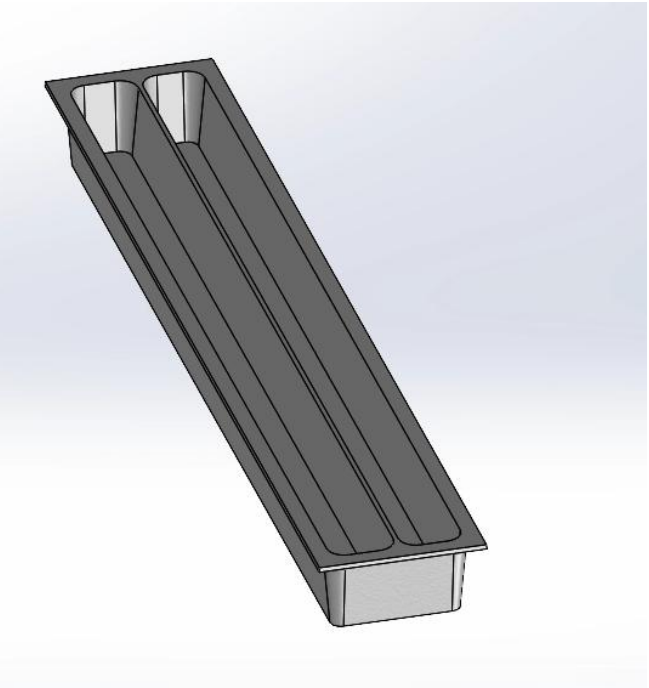
Shared Components Between Product A & B	Different Components between Product A & B
● Small Link ● Big Link ● Handles ● Flaps ● Supporting Trays ● Base	● Dividers

Product Family Design: FDT

Shared Functions between Product A & B	Functional Differences between Product A & B
● Organize Tools ● Store Tools ● Access tools ● Separate tools/hardware ● Display Labels ● Open housing ● Expand levels	● Adjust Configuration ● Prevent damage

Product Family Design: CAD Modules: Primary and Secondary Customers





Side By Side Comparison of Trays A & B Highlighting different compartments

Product Family Design: PFAD

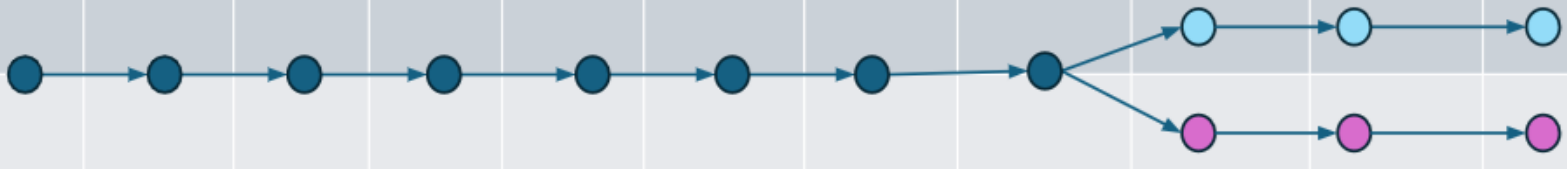
Original:

Variety	Base	Small Link 1	Large Link	Small Link 2	Tray 1	Tray 2	Divider 1	Divider 2	Flaps	Handles	Final Product
Product A											
Product B											



Improved:

Variety	Base	Tray 1	Tray 2	Flaps	Small Link 1	Small Link 2	Large Link	Handles	Divider 1	Divider 2	Final Product
Product A											
Product B											



Design For Manual Assembly: Part Selection

The manually oriented and assembled parts are our targets for redesign.

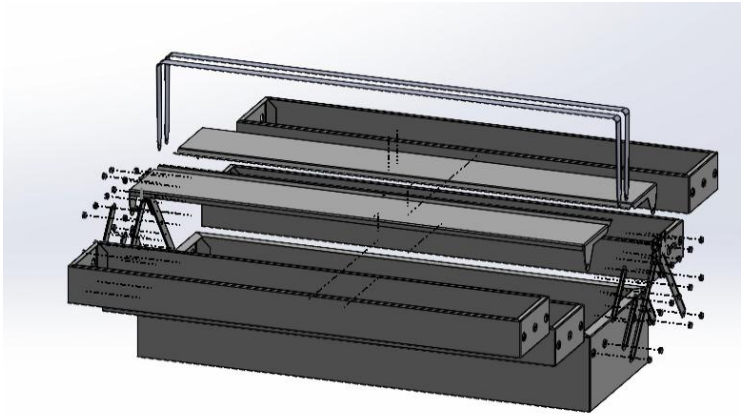
Part No.	Name	Quantity
1	Base	1
4	Supporting Tray	4
5	Divider A	2
6	Divider B	2
7	Flaps	2

Design For Manual Assembly: Before

DFA-6: Increase Part Symmetry

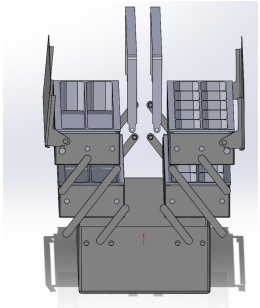
DFA-10: Adding features for easy insertion

DFA-12: Z Assembly



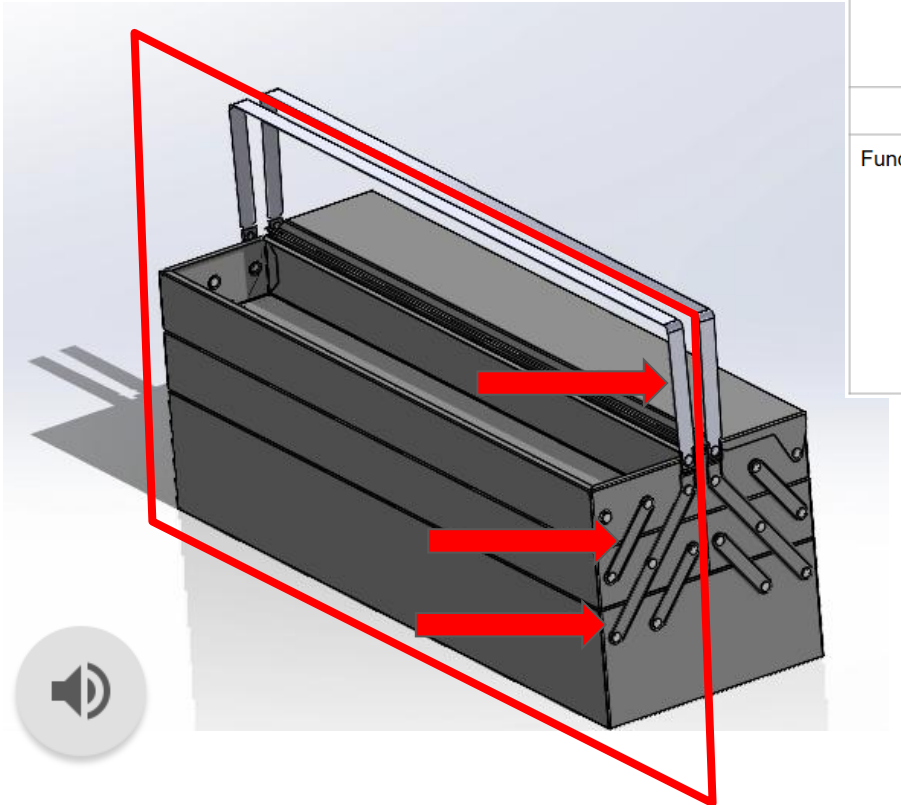
Item #	Name	a	b	Size	Number of items	Two-digit manual handling code	Manual handling time per part (sec)	Two-digit manual insertion code	Manal insertion time per part (sec)	Total assembly time	Theoretical minimum number of parts
1	Base	**360**	**180**	**5**	1	20	1.8	08	6.5	8.30	1
4	Supporting Tray	**360**	**180**	**31**	4	20	1.8	08	6.5	8.30	4
5	Divider A	**360**	**180**	**25**	2	20	1.8	08	6.5	8.30	2
6	Divider B	**360**	**180**	**7**	2	20	1.8	08	6.5	8.30	2
7	Flaps	**360**	**180**	**25**	2	20	1.8	08	6.5	8.30	2
total					11					41.50	11

Design For Manual Assembly: After



Item #	Name	a	b	Size	Number of items	Two-digit manual handling code	Manual handling time per part (sec)	Two-digit manual insertion code	Manal insertion time per part (sec)	Total assembly time	Theoretical minimum number of parts
1	Base	**360**	**180**	**5**	1	10	1.5	06	5.5	7.00	1
4	Supporting Tray	**360**	**180**	**31**	4	10	1.5	06	5.5	7.00	4
5	Divider A	**360**	**180**	**25**	2	10	1.5	06	5.5	7.00	2
6	Divider B	**360**	**180**	**7**	2	10	1.5	06	5.5	7.00	2
7	Flaps	**360**	**180**	**25**	2	10	1.5	06	5.5	7.00	2
total					11					35.00	11

Design For Automated Assembly: Subassembly Selection

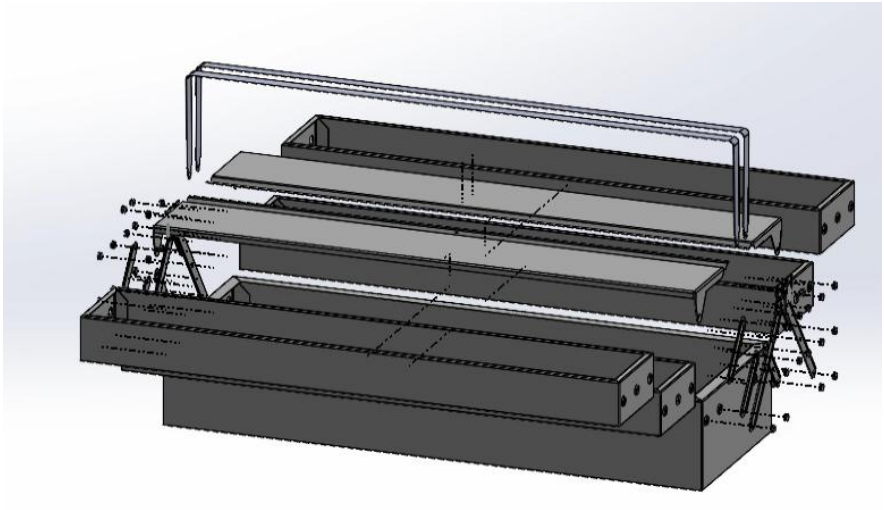


		Manufacturing Cost/Assembly	
		Low	High
Functional Importance	Low	- Fasteners - Flaps	
	High	- Base A/B - Handles	- Supporting Tray - Divider A - Divider B - Small Link - Big Link

"The" Quad Chart for DFM/A

		manufacturing cost	
		low	high
functional importance	low	consider elimination	consider elimination
	high	keep as it is	redesign with DFM/A

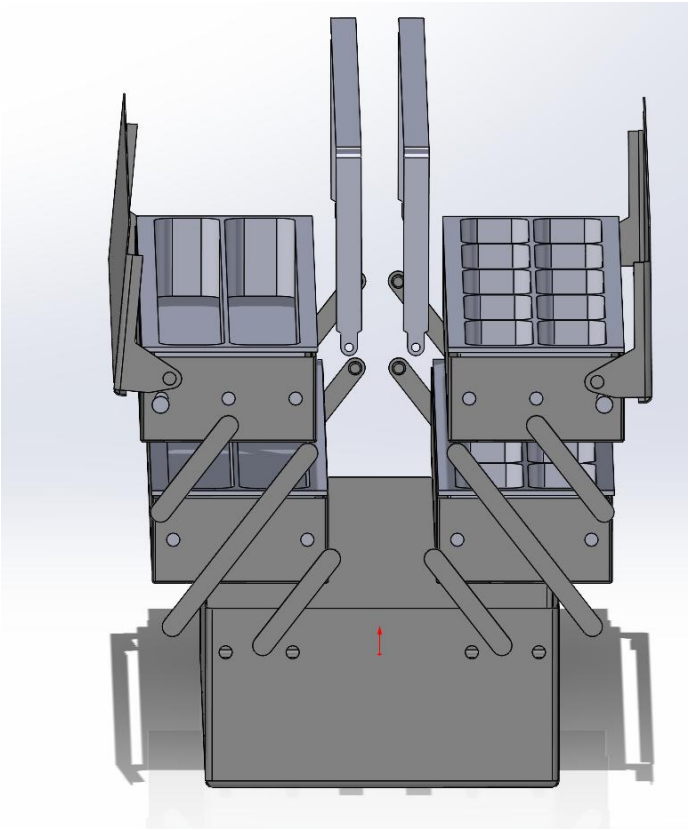
Design For Automated Assembly: Before



Item #	Name	# of Items	5-digit automatic handling code	Orienting efficiency	Relative handling cost	Max feed rate	handling (feeding) cost	2-digit automatic insertion code	Relative work head cost	Insertion cost	Total assembly cost	Theoretical min # of parts
1	Small Link	4	70040	0.9	3	112.5	0.048	18	2.1	3.15	12.79	4
2	Handle	1	72500	0.4	1	40	0.045	08	2	3	3.045	1
3	Big Link	2	70040	0.9	3	112.5	0.048	18	2.1	3.15	6.396	2
4	Screws	16	11000	0.3	1	37.5	0.048	49	2	3	48.77	0
Total											71	7

DFA Principles considered:
-DFA-1: Reduce Part count
-DFA-2: Reduce fasteners
-DFA-11: Provide locating features

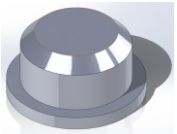
Design For Automated Assembly: After



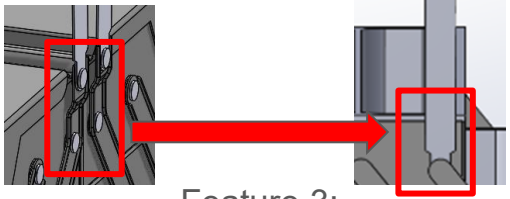
Item #	Name	# of Items	5-digit automatic handling code	Orienting efficiency	Relative handling cost	Max feed rate	handling (feeding) cost	2-digit automatic insertion code	Relative work head cost	Insertion cost	Total assembly cost	Theoretical min # of parts
1	Small Link	4	70000	0.9	1	112.5	0.016	10	1.2	1.8	7.264	4
2	Handle	1	72500	0.4	1	40	0.045	12	1.6	2.4	2.445	1
3	Big Link	2	70000	0.9	1	112.5	0.016	10	1.2	1.8	3.632	2
4	Riveting Operation	-	-	-	-	-	-	91	0.9	1.35	18.9	-
6												
Total											32.24	7



Feature 1:
Locating pins



Feature 2:
Riveting features
on pins



Feature 3:
Combine hole/rivet for handle
and big link

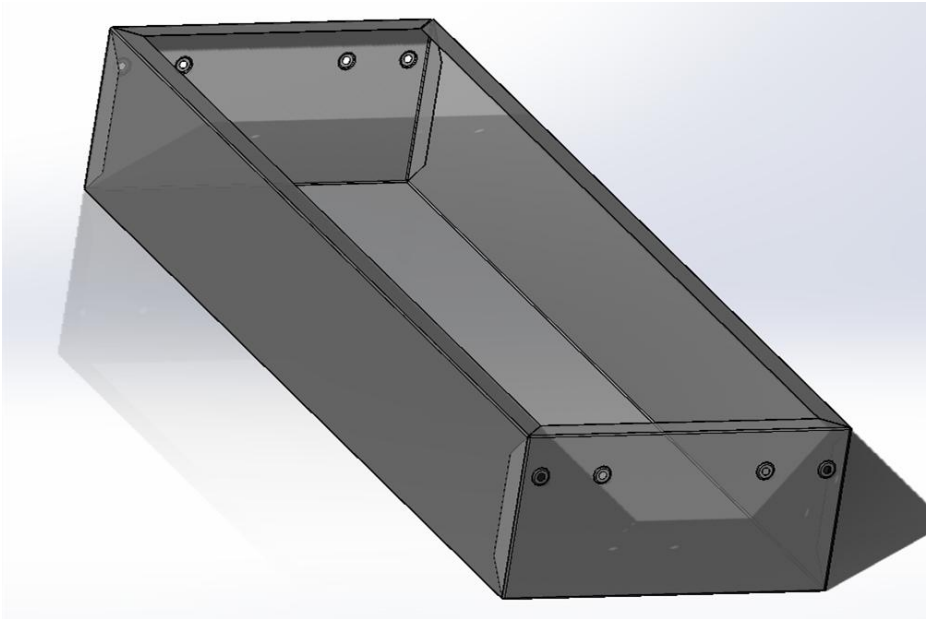
Design for Manufacturing: Process Selection

Aluminum Metal (non ferrous)	Folded Sheet Metal	1.64 kg	0.5 mm	0.1 mm	Not Strict	168 000
Process Compatible with Material	**Shape**	**Mass**	**Section Thickness**	**Tolerance**	**Surface Roughness**	**Economic Batch Size**
Sand Casting	Fails					
Die Casting	Fails					
Investment Casting	Fails					
Centrifugal Casting	Fails					
Forging	Fails					
Extrusion	Fails					
Sheet forming	Capable	Capable	Capable	Capable	Capable	Capable
Powder methods	Fails					
Electro-Machining	Capable	Capable	Capable	Fails		
Machining	Capable	Capable	Capable	Capable	Capable	Capable

Design for Manufacturing: Process Selection

Thermoplastics ABS	**Prismatic (non circular)**	**2.14kg**	**3 mm**	**0.5 mm**	**Not Strict**	**168 000**
Process Compatible with Material	**Shape**	**Mass**	**Section Thickness**	**Tolerance**	**Surface Roughness**	**Economic Batch Size**
Machining	Capable	Capable	Capable	Capable	Capable	Capable
Injection moulding	Capable	Capable	Capable	Capable	Capable	Capable
Blow moulding	Fails					
Compression moulding	Fails					
Rotational moulding	Fails					
Thermoforming	Fails					
Additive manufacture	Fails					

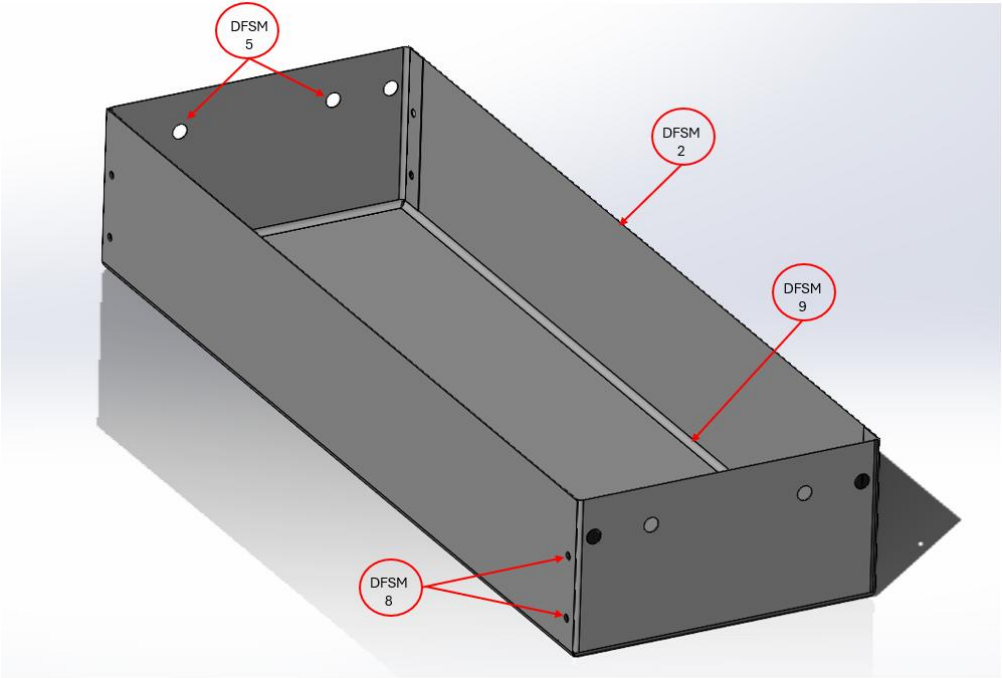
Design for Sheet Metal Manufacturing (Original)





Material Cost	$0.61/part
Operational Cost	$1.16/part
Mold Cost	$0.14/part
Total Cost per Part	**$1.91/part**

Design for Sheet Metal Manufacturing (Redesigned)



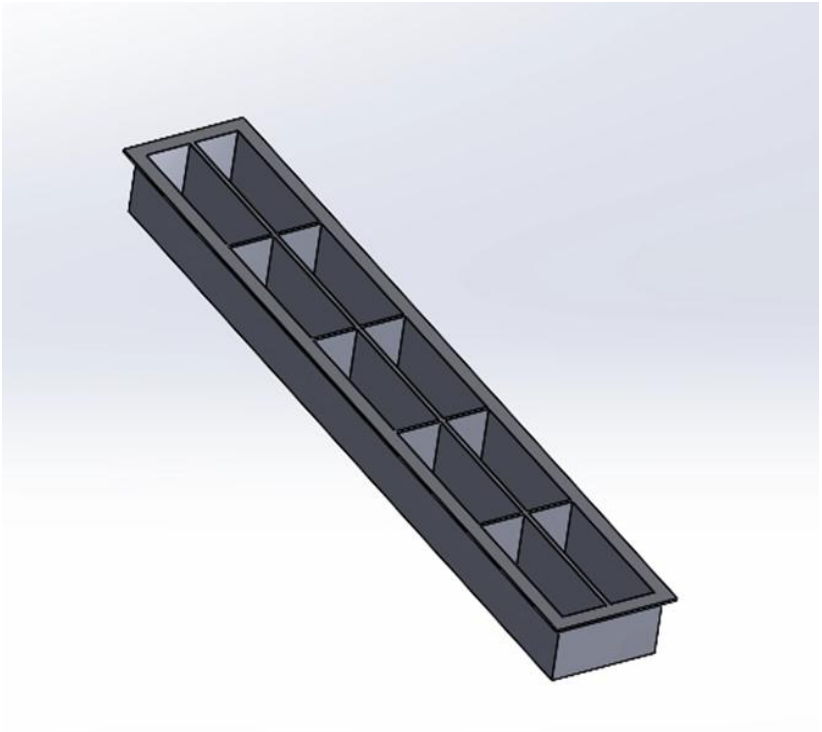
DFSM Principles Applied:

DFSM 2: Minimize setups and steps
DFSM 5: Allow the use of Standard Punches
DFSM 8: Avoid too small and close holes
DFSM 9: Avoid sharp bends



	Analytical	Hubbs
Original Part	$1.91	N/A
Redesigned Part	$1.67	$30.87
% Improvement	12.5%	Infinite

Design for Manufacturing
Tray 1 (Before)



Flat Parting Surface

Tolerance +/- 0.325 mm

Surface Finish: Not Critical

Length: 615 mm

Width: 105 mm

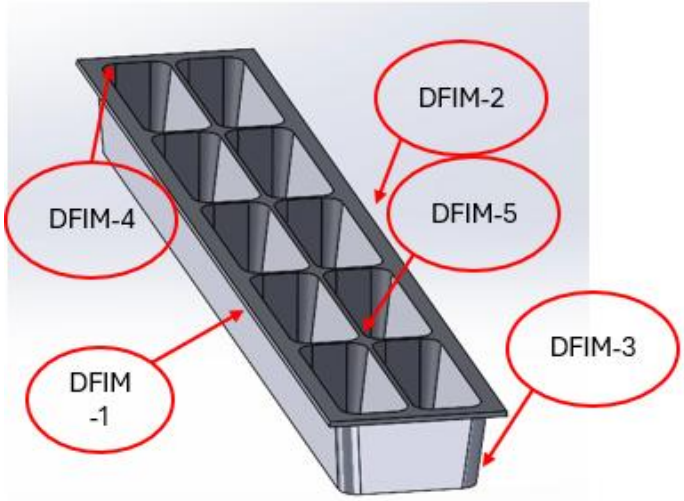
Volume: 649.39 cm^3

Maximum wall thickness: 5mm

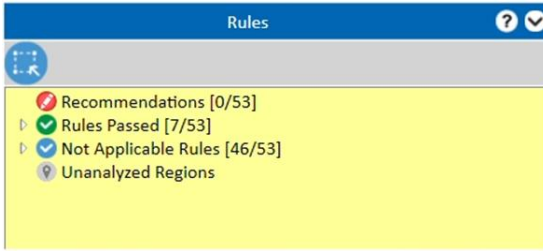


Tray 1	Cost Per Part
Material Cost	$2.13
Operating Cost	$1.06
Mold Cost	$0.06
Total Cost	$3.25

Design for Manufacturing
Tray 1 (After)



- DFIM-1: Reduce Wall Thickness
- DFIM-2: Uniform Wall Thickness
- DFIM-3: Add Draft Angles
- DFIM-4: Avoid Sharp Corners
- DFIM-5: Keep rib thickness 60% of wall thickness



Flat Parting Surface

Tolerance +/- 0.325 mm

Surface Finish: Not Critical

Length: 615 mm

Width: 105 mm

Volume: 556.40 cm^3

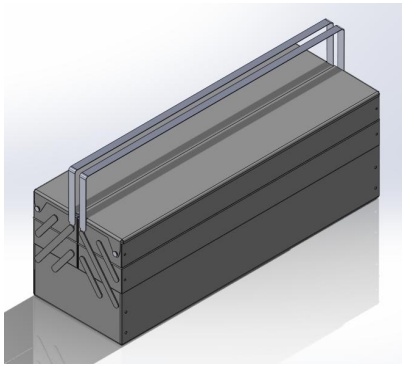
Wall thickness: 3mm

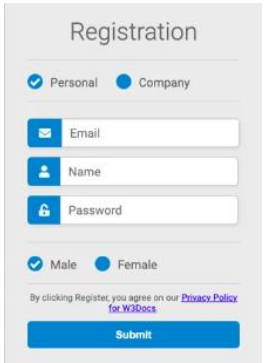
Tray 1 After	Cost Per Part
Material Cost	$1.82
Operating Cost	$0.48
Mold Cost	$0.06
Total Cost	$2.36

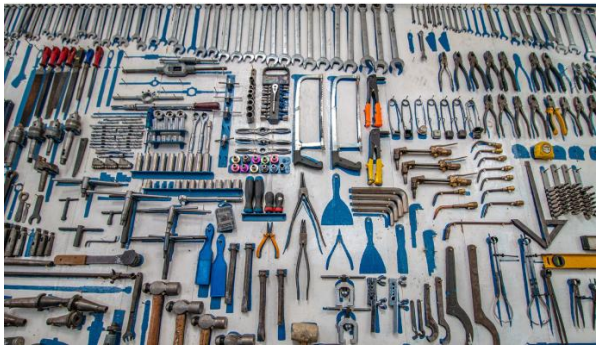
Product Service System

Borrow-a-tool Service:

- Offers members access to niche, job specific tools
- Users purchase our toolbox, register the product online, and create a borrow-a-tool account
- Members borrow tool for a period of time and return at no cost aside from membership fee
- Toolbox owner receive first year free







Conclusion/Design Critique

- Overall Design
  - Opportunity to add kickstands on the far end bases for balance & support
  - Modify link geometry to prevent upper-most tray from obstructing lower-most tray
- DFA (Automated)
  - Pro: cost reduction by ~$40
  - Con: Production volumes and majority of component sizes don’t justify automated assembly
- DFM (Base)
  - Pro: Simplified geometry and reduced cost by $0.24/part
  - Con: Not particularly aesthetic, still required fasteners to hold geometry
- DFM (Divider A&B)
  - Pro: Cost Reduction by ~$0.90 per part
  - Con: Minor loss in storage volume due to draft angles

